

MCTH10 4G LTE Cellular Temperature & Humidity Sensor Datasheet

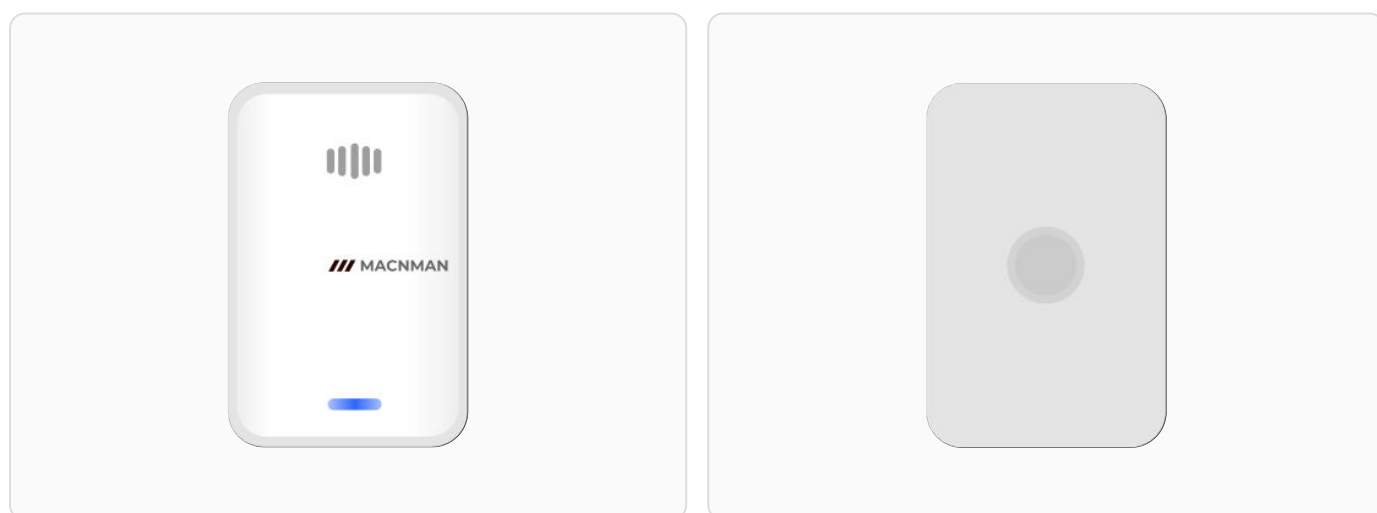
Product Overview

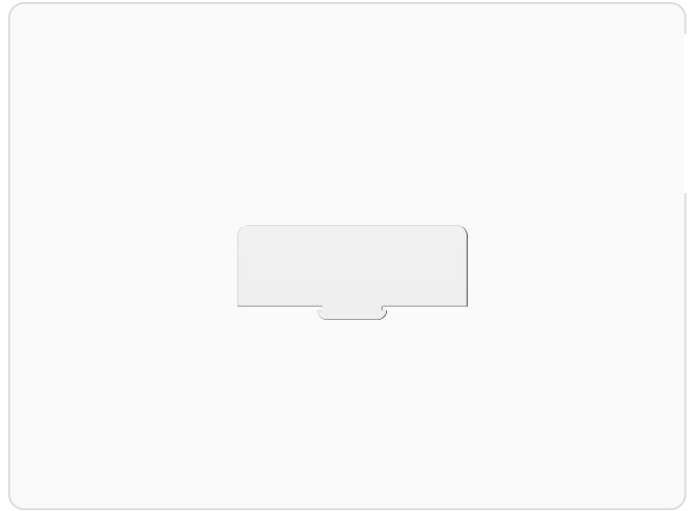
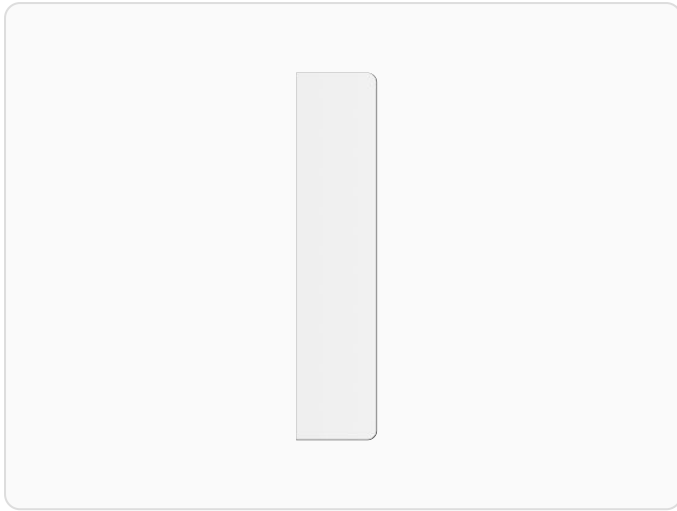
MCTH10 is an industrial-grade **Cellular Temperature & Humidity Monitoring Sensor** designed for continuous environmental monitoring in commercial and industrial infrastructure. Powered by a stable **12–24V DC supply**, the device delivers reliable 24/7 operation without battery limitations, making it ideal for permanent installations and mission-critical monitoring points.

MCTH10 measures **temperature and humidity with high accuracy** and transmits real-time data over **cellular connectivity**, eliminating the need for gateways, Wi-Fi networks, or local IT infrastructure. This makes it perfect for deployments across distributed sites such as **cold storage facilities, warehouses, pharma storage rooms, data centers, server rooms, greenhouses, and industrial buildings**.

With cloud connectivity, MCTH10 enables **live dashboards, instant alerts, automated compliance logging, and preventive maintenance reporting**, ensuring your operations stay protected from temperature excursions, humidity rise, and environmental instability.

Whether you need **remote monitoring, compliance readiness, or 24/7 environmental visibility**, **MCTH10 delivers always-on performance with cellular reliability and industrial-grade durability**.





Key Features

Cellular Temperature & Humidity Monitoring

Accurate real-time monitoring of temperature and humidity for industrial and commercial environments.

Direct-to-Cloud Connectivity

Sends sensor data over cellular network without requiring Wi-Fi, Ethernet, or any gateway infrastructure.

Live Dashboard & Data Logging

Continuous data capture with timestamped logs for trend analysis, audits, and compliance reporting.

Remote Configuration & Easy Deployment

Quick installation with remote device setup for reporting intervals, alert rules, and site configuration.

12–24V DC Powered Operation

Always-on performance with stable DC supply, ideal for fixed installations and critical monitoring locations.

Instant Alerts & Threshold Triggers

Configure high/low temperature and humidity limits to receive instant alerts for excursions and abnormal conditions.

Industrial-Grade Rugged Design

Built for long-term deployments in harsh environments with stable performance and durable enclosure.

Ideal for Cold Storage & Critical Rooms

Perfect for cold rooms, pharma storage, warehouses, data centers, and other temperature-sensitive operations.

Suited Applications

Cold Storage & Freezer Rooms

Continuous temperature and humidity monitoring to prevent spoilage, excursions, and compliance failures.

Warehouses & Logistics Facilities

Tracks environmental conditions across large storage zones to protect inventory and reduce losses.

Pharma & Vaccine Storage

Ensures stable storage conditions with real-time alerts and logging for audits and regulatory compliance.

Data Centers & Server Rooms

Monitors heat and humidity rise to prevent equipment damage and reduce unexpected downtime.

HVAC Monitoring & Utility Rooms

Helps validate HVAC performance and detect ventilation or cooling issues early.

Greenhouses & Indoor Farming

Maintains optimal temperature and humidity conditions for improved yield and crop consistency.

Industrial Plants & Production Areas

Supports stable environmental monitoring for sensitive processes, machines, and quality control zones.

Remote Sites Without Local Network

Ideal for locations where Wi-Fi/Ethernet is unavailable — cellular enables direct cloud connectivity.

Technical Specifications

General Specifications

Part Order Number	MCTH10
Product Brand	Macnman Technologies Pvt Ltd

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Temperature Measurement

Range	-30°C to +70°C
Accuracy	0°C to +70°C: ±0.3°C & -30°C to 0°C: ±0.6°C
Resolution	0.1°C
Calibration	Factory calibrated
Probe Options	Internal & External Both Options

Humidity Measurement

Range	0% to 100% RH
Accuracy	10% to 90% RH: ±3% RH & <10% or >90% RH: ±5% RH
Resolution	0.5% RH

Calibration	Factory calibrated
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Pressure Measurement*

Range	300 hPa to 1100 hPa
Accuracy	±1.0 hPa (typical at 25°C, 300–1100 hPa)
Resolution	0.1 hPa
Long-Term Stability	≤ ±1 hPa/year
Calibration	Factory calibrated

Wireless Transmission

Communication Module	Quectel EC200 Series
Network Technology	4G LTE (Cat 1) 2G GSM (Fallback)
Supported LTE Bands*	B1 / B3 / B5 / B8 / B40 / B41 (Region dependent)
Supported GSM Bands*	900 / 1800 MHz
Tx Power	LTE: Up to +23 dBm GSM: Up to +33 dBm
Receiver Sensitivity*	LTE: Up to -100 dBm (typical) GSM: Up to -109 dBm (typical)
Cloud Protocols	MQTT / HTTPS
Antenna	External high-efficiency cellular antenna
SIM Support	Nano SIM
Data Transmission	Configurable reporting interval with event-based alerts

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of device is being modified

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud Platforms

Supports Major Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub Custom Cloud via MQTT / HTTPS
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Physical Characteristics

Power Supply(Optional)	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP35 standard (IP67 optional variant)
Housing Material	Glass field Nylon
Color	Matte White
Weight	~95g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	12-24 V DC
Power Consumption (Typical)	Idle / Monitoring: < 10 mA Cellular Transmission Peak: Up to 500 mA
Protection	Reverse polarity protection Over-voltage protection Surge / transient protection

Communication Options

Cellular Connectivity	4G LTE Cat 1 with 2G fallback (via Quectel EC200)
BLE Connectivity	For local configuration and diagnostics via Macnman Maya App
Cloud Protocols	MQTT / HTTPS / TCP-IP
SIM Support	Nano SIM
Serial Interface (Optional)	RS485 Modbus RTU

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MCTH10 Variants

VARIANT NAME	POWER SOURCE	INPUT VOLTAGE	SOLAR SUPPORT	EXTERNAL FLASH	IDEAL FOR
MCTH10	External Power	12–24 V DC			Fixed installations with continuous monitoring and always-on cellular connectivity

What's in the Box?

Make sure box contains following items before installations



MLTH 30x



**Mounting
Clip***



**User manual &
Warranty Card**



**IP
Extender***



**Mounting
Screws**

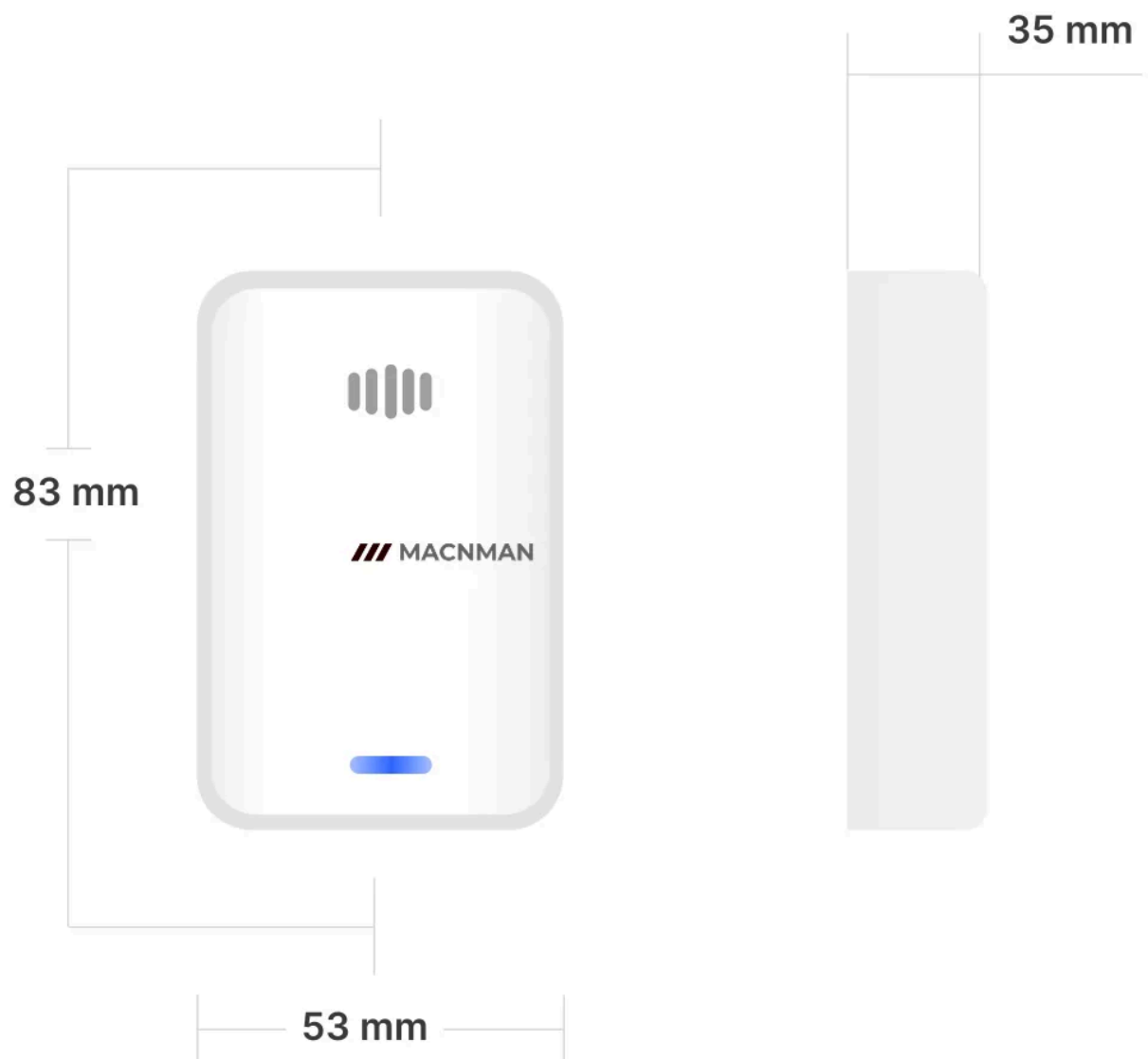


**Power
Adapters***

i Note

Accessories must be purchased separately & may not be included in the box.

MCTH10 Dimensions



Frequently Asked Questions

What is the MCTH10 used for?

MCTH10 is an industrial cellular temperature and humidity sensor for continuous environmental monitoring in commercial and industrial spaces over 4G LTE. Typical applications include Cold Storage & Freezer Rooms, Pharma & Vaccine Storage, Warehouses & Logistics Facilities, Data Centers & Server Rooms, HVAC Monitoring & Utility Rooms and Greenhouses & Indoor Farming.

What does the MCTH10 measure, and how accurately?

Measurement specifications of the MCTH10:

- Temperature: range -30°C to +70°C, accuracy 0°C to +70°C: $\pm 0.3^{\circ}\text{C}$ & -30°C to 0°C: $\pm 0.6^{\circ}\text{C}$, resolution 0.1°C.
- Humidity: range 0% to 100% RH, accuracy 10% to 90% RH: $\pm 3\%$ RH & <10% or >90% RH: $\pm 5\%$ RH, resolution 0.5% RH.
- Pressure: range 300 hPa to 1100 hPa, accuracy ± 1.0 hPa (typical at 25°C, 300–1100 hPa), resolution 0.1 hPa.

Which cellular networks does the MCTH10 support?

The MCTH10 uses a Quectel EC200 Series module for 4G LTE (Cat 1), 2G GSM (Fallback) connectivity. Supported LTE bands: B1 / B3 / B5 / B8 / B40 / B41 (Region dependent). Supported GSM bands: 900 / 1800 MHz. It takes a Nano SIM and sends data over MQTT / HTTPS.

What is the IP rating and operating temperature of the MCTH10?

The MCTH10 has an ingress protection rating of IP35 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MCTH10 configured?

The MCTH10 is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Related Products and Resources

- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)